

Questions with Answer Keys

MathonGo

Q1 - 2024 (27 Jan Shift 2)

The integral $\int \frac{(x^8 - x^2) dx}{(x^{12} + 3x^6 + 1) \tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$ equal to :

- (1) $\log_e \left(\left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| \right)^{1/3} + C$
- (2) $\log_e \left(\left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| \right)^{1/2} + C$
- (3) $\log_e \left(\left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| \right) + C$
- (4) $\log_e \left(\left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| \right)^3 + C$

Q2 - 2024 (29 Jan Shift 1)

For $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, if $y(x) = \int \frac{\operatorname{cosec} x + \sin x}{\operatorname{cosec} x \sec x + \tan x \sin^2 x} dx$ and $\lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} y(x) = 0$ then $y\left(\frac{\pi}{4}\right)$ is equal to

- (1) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$
- (2) $\frac{1}{2} \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$
- (3) $-\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$
- (4) $\frac{1}{\sqrt{2}} \tan^{-1}\left(-\frac{1}{2}\right)$

Q3 - 2024 (29 Jan Shift 2)

If

$$\int \frac{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x \sin(x-\theta)}} dx = A\sqrt{\cos \theta \tan x - \sin \theta} + B\sqrt{\cos \theta - \sin \theta \cot x} + C$$

where C is the integration constant, then AB is equal to

- (1) $4 \operatorname{cosec}(2\theta)$
- (2) $4 \sec \theta$
- (3) $2 \sec \theta$
- (4) $8 \operatorname{cosec}(2\theta)$

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Solutions

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Q1

$$I = \int \frac{x^8 - x^2}{(x^{12} + 3x^6 + 1) \tan^{-1}\left(x^3 + \frac{1}{x^3}\right)} dx$$

$$\text{Let } \tan^{-1}\left(x^3 + \frac{1}{x^3}\right) = t$$

$$\Rightarrow \frac{1}{1 + \left(x^3 + \frac{1}{x^3}\right)^2} \cdot \left(3x^2 - \frac{3}{x^4}\right) dx = dt$$

$$\Rightarrow \frac{x^6}{x^{12} + 3x^6 + 1} \cdot \frac{3x^6 - 3}{x^4} dx = dt$$

$$I = \frac{1}{3} \int \frac{dt}{t} = \frac{1}{3} \ln |t| + C$$

$$I = \frac{1}{3} \ln \left| \tan^{-1}\left(x^3 + \frac{1}{x^3}\right) \right| + C$$

$$I = \ln \left| \tan^{-1}\left(x^3 + \frac{1}{x^3}\right) \right|^{1/3} + C$$

Hence option (1) is correct

Q2

$$y(x) = \int \frac{(1 + \sin^2 x) \cos x}{1 + \sin^4 x} dx$$

$$\text{Put } \sin x = t$$

$$= \int \frac{1 + t^2}{t^4 + 1} dt = \frac{1}{\sqrt{2}} \tan^{-1} \frac{\left(t - \frac{1}{t}\right)}{\sqrt{2}} + C$$

$$x = \frac{\pi}{2}, t = 1 \quad \therefore C = 0$$

$$y\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{-1}{\sqrt{2}}\right)$$

Q3

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Solutions

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$$\int \frac{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x \sin(x - \theta)}} dx$$

$$I = \int \frac{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x (\sin x \cos \theta - \cos x \sin \theta)}} dx$$

$$= \int \frac{\sin^{\frac{3}{2}} x}{\sin^{\frac{3}{2}} x \cos^2 x \sqrt{\tan x \cos \theta - \sin \theta}} dx + \int \frac{\cos^{\frac{3}{2}} x}{\sin^2 x \cos^{\frac{3}{2}} x \sqrt{\cos \theta - \cot x \sin \theta}} dx =$$

$$\int \frac{\sec^2 x}{\sqrt{\tan x \cos \theta - \sin \theta}} dx + \int \frac{\operatorname{cosec}^2 x}{\sqrt{\cos \theta - \cot x \sin \theta}} dx$$

$$I = I_1 + I_2 \dots \dots \text{Let } \}$$

For I_1 , let $\tan x \cos \theta - \sin \theta = t^2$

$$\sec^2 x dx = \frac{2t dt}{\cos \theta}$$

For I_2 , let $\cos \theta - \cot x \sin \theta = z^2$

$$\operatorname{cosec}^2 x dx = \frac{2z dz}{\sin \theta}$$

$$I = I_1 + I_2$$

$$= \int \frac{2t dt}{\cos \theta} + \int \frac{2z dz}{\sin \theta}$$

$$= \frac{2t}{\cos \theta} + \frac{2z}{\sin \theta}$$

$$= 2 \sec \theta \sqrt{\tan x \cos \theta - \sin \theta} + 2 \operatorname{cosec} \theta \sqrt{\cos \theta - \cot x \sin \theta}$$

Comparing

$$AB = 8 \operatorname{cosec} 2\theta$$

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